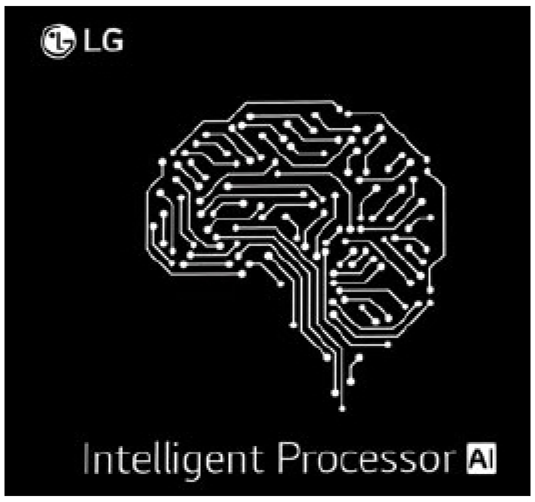
LG to accelerate development of artificial intelligence with own chip

As part of its strategy to accelerate the development of artificial intelligence (AI) devices for the home, LG has developed its own artificial intelligence chip (AI Chip) with proprietary LG Neural Engine to better mimic the neural network of the human brain to greatly improve processing of deep learning algorithms.

AI Chip incorporates visual intelligence to better recognise and distinguish space, location, objects and users, while voice intelligence accurately recognises voice and noise characteristics. Product intelligence enhances the capabilities of the device by detecting physical and chemical changes in the environment. The chip also makes it possible to implement customised AI services by processing and learning from video and audio data to enhance recognition of the user's emotions and behaviours, and situational context.



(Credit: www.lg.com)

Products utilising LG AI Chip take advantage of on-device AI to operate even without a network connection. The chip employs a powerful security engine to better protect personal data from external hacking. Processing that does not require high security is designed to run in a general zone, and jobs that require higher security run in a separate hardware-implemented security zone.

LG's AI Chip is designed to enhance recognition performance in the products it powers, such as an advanced image-recognition engine for simultaneous localisation and mapping (SLAM). For example, its powerful image-processing function corrects for distortion caused by wide-angle lenses, and generates brighter and clearer images in dark environments.

LG Electronics plans to include the new AI Chip in future robot vacuum cleaners, washing machines, refrigerators and even air-conditioners. In addition, LG will expand the reach of its AI solutions through collaborations with outside companies, universities and research laboratories.

Self-driving truck makes the first commercial cross-country trip

A Silicon Valley startup has completed what appears to be the first commercial freight cross-country trip by an autonomous truck, which finished a 4506.2km (2800-mile) run from Tulare, California to Quakertown, Pennsylvania for Land O'Lakes in under three days.

According to Plus.ai, a three-year-old company in Cupertino, that was responsible for the feat, a safety driver was aboard the autonomous semi, ready to take the wheel if needed, along with a safety engineer who observed how things were going.

"We wanted to demonstrate the safety, reliability and maturity of our overall system," says Shawn Kerrigan, co-founder and chief operating officer of Plus.ai. The system uses cameras, radar and lidar—laser-based technology to help vehicles determine distance—and handles different terrains and weather conditions such as rain and low visibility well.

The truck, which travelled on Interstates 15 and 70 right before Thanksgiving, had to take scheduled breaks but drove mostly autonomously. There were zero disengagements, or times the self-driving system had to be suspended because of a problem, according to Kerrigan.

End of year is peak butter time, according to Land O'Lakes. "To be able to address this peak demand with a fuel- and cost-effective freight transport solution will be tremendously valuable to our business," says Yone Dewberry, the butter maker's chief supply officer.

How long will it be before self-driving trucks are delivering goods regularly across the nation's highways? Kerrigan says it is "a few years out."

Singapore hotels use facial recognition for checking in guests

Tourists visiting Singapore can now check in at some hotels using facial recognition technology under a pilot programme that could cut waiting times and help tackle a labour crunch.

The country of 5.7 million people is increasingly turning to automation to speed up services and deal with workforce shortages, with robots deployed for tasks ranging from cleaning to making noodles.

Under the pilot, visitors will not need to wait to be checked in by hotel staff but can instead use a phone app fitted with facial recognition technology or machines that scan their passports. Data from the scan will be sent to immigration authorities for checks after which visitors will be issued a room key, according to Singapore Tourism Board and Hotel Association.

The technology, reportedly being trialled at three hotels, is similar to that used in some airports including Singapore's Changi. It could reduce check-in times by up to seventy per cent.